

Exercise 1

1.1 Homogeneous Coordinates

$$a) \begin{cases} \hat{I}_1 = (a_1, b_1, c_1)^T \\ \hat{I}_2 = (a_2, b_2, c_2)^T \end{cases} \begin{cases} a_1x + b_1y + c_1 = 0 \\ a_2x + b_2y + c_2 = 0 \end{cases} \quad \hat{X} = \hat{I}_1 \times \hat{I}_2 = (b_1c_2 - b_2c_1, a_2c_1 - a_1c_2, a_1b_2 - b_1a_2)^T$$

$$b) \begin{cases} \hat{X}_1 = (a_1, b_1, c_1)^T \\ \hat{X}_2 = (a_2, b_2, c_2)^T \end{cases} \quad \hat{I} = \hat{X}_1 \times \hat{X}_2 = (b_1c_2 - b_2c_1, a_2c_1 - a_1c_2, a_1b_2 - b_1a_2)^T$$

$$c) \begin{cases} x + y + 3 = 0 \\ -x - 2y + 7 = 0 \end{cases} \Rightarrow \begin{cases} \hat{I}_1 = (1, 1, 3)^T \\ \hat{I}_2 = (-1, -2, 7)^T \end{cases} \quad \hat{X} = \hat{I}_1 \times \hat{I}_2 = (13, 10, -1) = (-13, -10, 1)$$

$$d) \begin{pmatrix} 3 \\ 5 \\ 1 \\ 3 \end{pmatrix}$$

$$e) \frac{1}{5} \begin{pmatrix} 2 \\ 14 \\ 69 \end{pmatrix}$$

Transformations

$$a) \begin{cases} 1 + x = 0 \\ 2 + y = 3 \end{cases} \quad \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$b) T = \begin{pmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \end{pmatrix} \quad E(t_1, t_2) = \sum_{i=1}^N (x_i^2 + t_1 - y_i^2 + t_2 - y_i^2)$$

$$\frac{\partial E}{\partial t_1} = \sum_{i=1}^N 2(x_i - y_i) + 2Nt_1 \quad \frac{\partial E}{\partial t_2} = \sum_{i=1}^N 2(x_i^2 - y_i^2) + 2Nt_2$$

$$t_1 = \frac{\sum_{i=1}^N (y_i - x_i)}{N} \quad t_2 = \frac{\sum_{i=1}^N (y_i^2 - x_i^2)}{N}$$

$$c) T = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 4 \end{pmatrix}$$

Camera Projections

$$a) R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \frac{\pi}{2} & -\sin \frac{\pi}{2} \\ 0 & \sin \frac{\pi}{2} & \cos \frac{\pi}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \quad \begin{pmatrix} R & t \\ 0^T & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad K = \begin{pmatrix} 100 & 0 & 25 \\ 0 & 100 & 25 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\hat{P} = \begin{pmatrix} K & 0 \\ 0^T & 1 \end{pmatrix} \begin{pmatrix} R & t \\ 0^T & 1 \end{pmatrix} = \begin{pmatrix} 100 & 25 & 0 & 150 \\ 0 & 25 & -100 & 50 \\ 0 & 1 & 0 & 2 \end{pmatrix}$$

$$b) \hat{P}^{-1} = \begin{pmatrix} \frac{1}{100} & 0 & -\frac{1}{4} & -\frac{1}{2} \\ 0 & \frac{1}{100} & \frac{1}{4} & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad \tilde{X}_w = \hat{P}^{-1} \tilde{X}_s = \begin{pmatrix} -\frac{1}{4} \\ \frac{1}{2} \\ -\frac{1}{4} \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} \quad X_w = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

c) i)

$$\text{corners } (10, 10, 5)^T (110, 10, 5)^T (-10, 10, 5)^T (-10, 10, 5)^T (10, 10, 25)^T (110, 10, 25)^T (-10, 10, 25)^T (-10, 10, 25)^T$$

$$K = \begin{pmatrix} 5 & 0 & 10 \\ 0 & 5 & 10 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{projected points } (20, 20)^T (20, 0)^T (0, 0)^T (0, 20)^T (12, 12)^T (12, 8)^T (8, 8)^T (8, 12)^T$$

ii)

$$K = \begin{pmatrix} 10 & 0 & 10 \\ 0 & 10 & 10 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{corners } (10, 10, 25)^T (110, 10, 25)^T (-10, 10, 25)^T (-10, 10, 25)^T (10, 10, 45)^T (110, 10, 45)^T (-10, 10, 45)^T (-10, 10, 45)^T$$

$$\text{projected points } (14, 14)^T (14, 10)^T (10, 10)^T (10, 14)^T (110, 110)^T (110, 70)^T (70, 70)^T (70, 110)^T$$

ii) corners $K = \begin{pmatrix} 0 & 0 & 10 \\ 0 & 0 & 10 \\ 0 & 0 & 1 \end{pmatrix}$

projected points $(10, 10, 10)^T$ $(10, -10, 10)^T$ $(-10, -10, 10)^T$ $(-10, 10, 10)^T$ $(10, 10, 30)^T$ $(10, -10, 30)^T$ $(-10, -10, 30)^T$ $(-10, 10, 30)^T$

$(20, 20)^T$ $(20, 0)^T$ $(0, 0)^T$ $(0, 20)^T$ $(\frac{40}{3}, \frac{40}{3})^T$ $(\frac{40}{3}, \frac{20}{3})^T$ $(\frac{20}{3}, \frac{20}{3})^T$ $(\frac{20}{3}, \frac{40}{3})^T$

iii) corners

$$K = \begin{pmatrix} 90 & 0 & 10 \\ 0 & 90 & 10 \\ 0 & 0 & 1 \end{pmatrix}$$

corners $(10, 10, 90)^T$ $(10, -10, 90)^T$ $(-10, -10, 90)^T$ $(-10, 10, 90)^T$ $(10, 10, 110)^T$ $(10, -10, 110)^T$ $(-10, -10, 110)^T$ $(-10, 10, 110)^T$

projected points $(20, 70)^T$ $(20, 0)^T$ $(0, 0)^T$ $(0, 20)^T$ $(\frac{200}{11}, \frac{200}{11})^T$ $(\frac{200}{11}, \frac{20}{11})^T$ $(\frac{20}{11}, \frac{20}{11})^T$ $(\frac{20}{11}, \frac{200}{11})^T$

iiii) $(0, 0)^T$ $(0, 20)^T$ $(20, 20)^T$ $(20, 0)^T$

iiii) distance $\rightarrow \infty$

Photometric Image Formation

a) $\frac{1}{Z_s} + \frac{1}{Z_c} = \frac{1}{f}$ $Z_c = 5100 \text{ mm}$

b) $l = \frac{d}{Z_s} \Delta Z_s = \frac{f}{N Z_s} \Delta Z_s$

c) $l_1 = 0.0625 \text{ mm}$ $l_2 = 0.01875 \text{ mm}$

$S_p = J_{ap} = 0.02 \text{ mm}$